

Dr Benya Wang

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CAREER PROFILE

Machine Learning Engineer with a PhD and 8 years of experience delivering enterprise-grade AI and analytics platforms across mining and resources. Proven track record owning the full ML lifecycle from architecture design through to production deployment and governance at industrial scale. Deep expertise in MLOps, LLM / GenAI pipelines, Vision AI, and edge computing on Azure, Databricks, and AWS. Translates complex business problems into scalable AI solutions through reusable governance frameworks and cross-functional adoption.

CORE COMPETENCIES

Cloud & Data Platforms

Databricks (data warehouse, ETL/ELT pipelines, workflow orchestration), Azure (AI Foundry, Azure ML, IoT Edge/Hub, Azure OpenAI, DevOps), AWS

MLOps & DevOps

MLflow, Azure Pipelines, GitHub Actions, Terraform, Bicep, Docker, CI/CD pipeline design

LLMs & Generative AI

Azure AI Foundry, Claude Code, LangChain, LightRAG, Microsoft Copilot Studio, RAG pipeline design, multi-agent orchestration

ML & Data Engineering

XGBoost, LightGBM, Python, PySpark, Pandas, SQL, Power BI

Leadership & Architecture

Enterprise platform strategy, AI governance framework design, stakeholder engagement, data governance, technical mentoring

PROFESSIONAL EXPERIENCE

Specialist Machine Learning Engineer

Apr 2023 – Present

South32 · Perth, WA

- Pioneered an air-gapped edge ML deployment pattern (Azure IoT Edge, ACR, Databricks Apps) enabling model inference on isolated industrial sites with zero internet connectivity; adopted as the enterprise-wide standard across all remote operations.
- Defined the technical strategy and built a unified AI platform (Azure AI Search, Azure OpenAI, Cosmos DB) that reduced time-to-production for new LLM applications by ~60%, enabling multiple product teams to ship independently.
- Led technical governance for the ML platform including authoring standards, reviewing architectures, and mentoring data scientists across business units to drive consistent, production-quality delivery.
- Delivered GenAI-powered document intelligence (LightRAG, Azure AI Foundry) across geoscience and safety corpora, surfacing actionable insights from previously unstructured data at enterprise scale.
- Partnered with Cyber and Enterprise Architecture teams to design a Microsoft Copilot Studio governance framework and a GitHub / GitHub Copilot governance framework, establishing enterprise-wide standards for AI-assisted development, access controls, and compliant deployment; supported the organisational migration from Azure DevOps to GitHub at scale.
- Served as Acting Data Science Lead May–Jun 2026, maintaining delivery continuity across active AI platform workstreams; co-designed the platform production support model with the Operational Digital Technology team, defining ownership, escalation paths, and SLA structures.

Specialist Data Scientist

Jan 2022 – Apr 2023

South32 · Perth, WA

- Architected and led delivery of a centralised Data Science platform (Azure Databricks, MLflow, Azure Pipelines) that standardised MLOps workflows across 6+ business units, cutting the average model deployment cycle from ~3 weeks to under 5 days and eliminating ad-hoc deployment risk.
- Led advanced feature engineering and exploratory analysis at scale using PySpark and scikit-learn across large industrial datasets.
- Partnered with data governance teams to implement stewardship and metadata management aligned with enterprise compliance policies.
- Designed and deployed a Vision AI safety system (YOLO, Supervision) for automated stop-sign monitoring, directly improving operational safety compliance.
- Implemented production multi-agent LLM pipelines (GraphRAG, AutoGen) to extract structured insights from HR records, safety incident reports, and geoscience exploration corpora.

Data Scientist

Jan 2021 – Jan 2022

Rio Tinto · Perth, WA

- Engineered serverless data pipelines (AWS Lambda, Glue, Athena) for real-time production analytics supporting operational decision-making.
- Led ML-driven optimisation of ore blending strategies, improving resource utilisation and reducing annual operational costs.
- Won an internal Hackathon for diamond image classification using TensorFlow and transfer learning.

Graduate Data Scientist

Feb 2019 – Jan 2021

Rio Tinto · Perth, WA

- Developed predictive maintenance models (LightGBM, Random Forest) for a 1,700 km rail network, reducing unplanned downtime through early fault detection.
- Containerised models in Docker and deployed via AWS SageMaker, establishing repeatable deployment and rollback processes for the team.
- Delivered spatial analytics dashboards (GIS maps, Plotly Dash) enabling operational teams to make data-driven decisions in real time.

Data Science Intern

Jan 2018 – Apr 2018

Rio Tinto · Perth, WA

- Contributed to early-stage predictive maintenance analytics and sensor data preprocessing pipelines.

EDUCATION

PhD in Environmental Engineering

Feb 2015 – Feb 2019

University of Western Australia

Thesis: Application of machine learning algorithms in urban catchments. Five peer-reviewed publications in *Geoscientific Model Development*, *Water Resources Research*, and other journals.

Bachelor of Science

Sep 2010 – Jun 2014

Zhejiang University, China

CERTIFICATIONS

Databricks Certified

Machine Learning Professional · Machine Learning Associate · Data Engineer Associate

Microsoft Certified

DevOps Engineer Expert · Azure Data Scientist Associate · Azure Data Engineer Associate · Azure Developer Associate

AWS Certified

Machine Learning – Specialty · Solutions Architect – Associate

Other

HashiCorp Certified: Terraform Associate · Anthropic: Claude Code in Action